

From Your Pocket to Your Boss's: Cash Transfers & Wage Theft

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Abstract

Wage theft is a pervasive issue for American workers, yet research on the subject is stymied by underreporting: workers fear retaliatory firing if they report. This paper investigates if reporting decisions are affected by work-contingent government transfers, as they should make remaining employed more attractive, increasing the cost of being fired. Using variation in state Earned Income Tax Credits over time, I find that increases in transfers are associated with lower reporting counts, analyzing at the ZIP code level. To confirm that this decrease truly reflects changes in reporting rather than a decrease in wage theft itself, I estimate the relationship between transfers and several intensive-margin outcomes of wage theft. Finding no evidence that transfers decrease wage theft intensity, I conclude that transfers reduce reporting propensities.

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I Introduction

Wage theft is a pervasive issue for American workers, wherein employers avoid paying workers their legally due wages. This can take many forms, such as failing to meet the minimum wage or withholding overtime pay. While it is known to be a widespread issue (costing workers an estimated \$50 billion annually) it is not well-studied due to underreporting — workers who report wage theft are often fired in retaliation, creating a significant deterrent (Meixell and Eisenbrey 2014).

To better understand how workers make reporting decisions, I explore the role of work-contingent government transfers. By increasing a worker’s total income, these transfers should increase the value of work and make potential retaliatory firing even more costly. To study this relationship, I propose a model in which workers weigh the potential gains from reporting (i.e., backwages repaid) against the potential costs (i.e., retaliatory firing). Simple comparative statics suggest that an increase in government transfers should decrease reporting propensities, as the loss from being fired outweighs the potential gains from a successful report.

Taking this model to the data on wage theft (documented by the Federal Wage & Hour Division) this prediction is borne out. As a measure of work-contingent transfers, I use differences in Earned Income Tax Credit (EITC) generosity. States match the federal EITC at varying levels, and many have adjusted those rates over time, producing plausibly exogenous variation at the state-year. Analyzing this relationship at the ZIP-code level via Poisson regression, I find that an increase in state EITC of 10 percentage points is associated with a decrease in ZIP wage theft reports of 4.4%. Scaling that up to the state level, this implies roughly 26 fewer reports per year (off an average of 600) and an additional \$286,000 in unreported stolen wages.

Wage theft is a complex challenge for American workers. Most estimates of its scale rely on small surveys, and what is formally documented only reflects those cases where the victims have chosen to come forward (Cooper and Kroeger 2017). It takes a variety of forms, particularly in industries with more complex wage structures such as tips or productivity

bonuses. Wage theft is also perpetrated by a diversity of employers, from large firms like Amazon or Bank of America to small businesses with just two employees (Sainato 2023). The issue has been of accordingly high interest in the economics literature, including studies of what factors make firms more likely to engage in theft (Ji and Weil 2010; Clemens 2021) and how the government can best deter this behavior (Weil 2010, 2018; Stansbury 2021).

When workers are victims of wage theft, their primary recourse is to file a report to the Wage & Hour Division of the Department of Labor.¹ This will usually lead to an investigation by the Division into what labor laws have been violated, and what backwages (or possible fines or damages) are owed. By one estimate, roughly \$1 billion is recovered each year - accounting for a minute fraction of the estimated theft (Meixell and Eisenbrey 2014).

Unfortunately, due to underreporting there is still much that is not understood about wage theft behavior, in particular as it interfaces with the welfare state and government transfers. The crime of wage theft falls most heavily on workers in low-wage industries who are also likely to rely on welfare and transfer programs, making the interaction of the two important for optimizing policy. The lack of research in this area was highlighted by a recent review article, Marinescu and Rosenfeld (2022), in the context of diminishing worker power.

Across several empirical specifications, I confirm a negative relationship between transfers and wage theft reporting; however, it is not clear *a priori* that this finding reflects a direct effect of transfers on worker reporting behavior itself. Reported wage theft is a product of both the propensity of workers to report and the propensities of employers to engage in the underlying wage theft. It could be the case that changes in EITC rates are actually affecting this second factor, the intensity of wage theft. This would result in a lower reporting frequency mechanically, as there is less real theft for workers to report. In other words, the negative relationship between transfers and observed reports could be driven by a mechanism of worker behavior (lowering their reporting propensities), or through a mechanism of employer behavior (lowering their wage theft propensities).

¹In most states, workers also have the opportunity to file a complaint with their state agency; however, the data on these state-level reports is heterogeneous and ill-maintained, making analysis of it highly challenging. Workers could also sue their employer directly in court, though this is often expensive and a lengthy process.

To understand this alternative mechanism, one can imagine an employer who sees the increased transfer payments as insulating their employee against the costs of being unemployed, thus diminishing the potency of the employer’s retaliatory firing threat. Such an employer would likely decrease their wage theft behavior in response to rising transfers, in order to forestall reports from their newly flush workers, which could be costly to the employer.

In order to differentiate between these two mechanisms, worker and employer behavior changes, I analyze the effect of transfers on secondary outcomes which I believe proxy for employer behavior. Going beyond the reporting frequency itself, these measures tell us something about the nature of the wage theft being reported. Essentially, if the observed decrease in reporting is being driven by employer behavior, one would expect that these secondary measures would also show some adjustment. Specifically, I use four metrics of employer wage theft: the number of ”empty” reports (where no theft or violations are found), the number of repeat-offenders reported, the number of legal violations found during investigations of reports, and the amount of civil monetary penalties assessed per report.

Across these outcomes, I fail to find evidence of reduced wage theft intensity or changes in behavior on the part of employers. While individually each of these secondary outcomes is not totally compelling, I do believe that in combination they are convincing that the decrease in reporting is not driven by the employer mechanism, but rather via my hypothesized mechanism of decreased worker reporting propensities.

This paper contributes to a growing literature on workplace crime and worker reporting incentives. As noted above, the observed data on various employer violations depend jointly on the true intensity of those violations and on the likelihood that violations are converted into documented reports (Marinescu et al. 2020). The importance of this distinction was made by companion papers Boone and Ours (2006) and Boone et al. (2011) in the context of workplace safety. While Boone and Ours (2006) show that empirically it appears workplace safety violations fall during recessions, Boone et al. (2011) clarified that the effect is largely driven by changes in reporting behavior, rather than actual workplace safety changes.

Boone et al., go further to offer a clear mechanism for the effect of recessionary shocks on reporting. Workers are primarily deterred from making reports of workplace crime by the threat of retaliatory firing. The decision to report or not must weigh the potential costs of being fired and becoming unemployed against remaining quiet and staying employed. This comparison of unemployment conditions to continued employment naturally depends on the outside unemployment rate — as a recession decreases the outside option of workers if they should become unemployed, the risk of a retaliatory firing for reporting workplace safety violations grows more costly, thus reports fall. A similar mechanism has been documented in a variety of different contexts: Hispanic workers facing increased threats of immigration enforcement report fewer safety violations (Grittner and Johnson 2022), low-income workers with fewer outside options are less comfortable discussing workplace issues (Hertel-Fernandez 2020), and reductions in unemployment benefits exacerbate underreporting of sexual harassment in the workplace (Dahl and Knepper 2021).

Across these settings, authors focus on the worsening unemployment situation as driving the mechanism. By a similar logic, an improvement in the condition of remaining employed should also decrease reporting, as the cost of becoming unemployed increases for the worker. This is the approach I take in this paper, using increases in the work-contingent transfers as an increase in the attractiveness of employment. While I focus on the crime of wage theft, I believe that a similar complementary mechanism likely exists for the other kinds of workplace crime which are studied in the literature.

The remainder of this article proceeds as follows: Section II reviews a theoretical model of wage theft reporting and its testable implications. Section III summarizes the data available for testing those implications. Section IV reviews the empirical strategy employed, and Section V its results. Section VI discusses these results and their implications, before Section VII concludes.

II Theory

To understand why wage theft is often not reported by its victims, it is helpful to consider the problem through the lens of a simple model. This section walks through one such framework² which predicts that increased transfer generosity will decrease reporting probabilities. The intuition is that transfers increase the value of employment relative to unemployment, such that the threat of being fired becomes sufficiently costly to outweigh the potential gains from reporting.

Consider an environment with a representative worker facing some exogenous amount of wage theft, $t \geq 0$. I abstract from the employer side of this relationship for simplicity. The worker, faced with a fixed wage w and theft t , must choose whether or not to report that theft to the authorities.

If the worker chooses not to report the theft, they will receive (with certainty) their wage net of theft and net of taxes τ , represented as $(1 + \tau)(w - t)$. Note that in this setup τ will be assumed positive, reflecting a situation similar to the real-world context of the Earned Income Tax Credit. This seems appropriate given our understanding of wage theft as a primarily low-wage phenomenon. For the predictions of this model, it is not important that τ be positive, only that the changes in tax structure result in increased net income for the worker.

If the worker chooses to report the theft, their outcome is less certain. A report will trigger an investigation into the situation by the government, which will succeed with a exogenous probability p . This uncertainty reflects the administrative challenges in proving wage theft, as employers may be able to evade detection. A successful report results in repayment of backwages, equal to the amount taken t , plus potentially additional damages, represented as γ . An unsuccessful report returns nothing for the worker.

Simultaneously, a report creates the risk of a retaliatory firing by the firm, occurring

²This model is based off that of Boone and Ours (2006) for the similar reporting issue of workplace safety, which has also been similarly applied in Dahl and Knepper (2021) for sexual assault reporting. My modification is to introduce tax rates.

with probability θ . A fired worker will either find a replacement job immediately (at new wage level w'), or they will remain unemployed and earn welfare benefits b . These events occur with probability q and $1 - q$, respectively. Accordingly, the value for the worker of not being fired, V_{NF} is their net wage per the success of their report (and net of taxes), $(1 + \tau)(w - (1 - p)t)$. The value for the worker of being fired, V_F is the probability-weighted sum of re-employment and unemployment, or $q(1 + \tau)w' + (1 - q)b$.

Formalizing this setup, we can calculate the expected value for the worker of making a report, and consider when it may outweigh the welfare value of not making a report:

$$p\gamma + \theta V_F + (1 - \theta)V_{NF} \geq (1 + \tau)(w - t) \quad (1)$$

The first term represents the potential payout of the report, the next two terms the uncertainty of potential retaliatory firing, and the right-hand side the status quo. This can be rewritten to compare the expected gains from reporting on one side to the expected losses on the other:

$$p\gamma + (1 - \theta)(1 + \tau)pt \geq \theta[q(1 + \tau)w' + (1 - q)b + (1 + \tau)(w - t)] \quad (2)$$

With this simplification, note that the gains on the left-hand side are increasing with respect to the amount of theft t , which is to say that the potential gains from choosing to report are increasing in theft. In contrast, the right-hand side is decreasing in t , as greater wage theft reduces the take home pay of the worker, and thus their potential losses from remaining employed and silent. Combining these two properties, it is clear that there must also therefore exist a level of theft, t^* such that this condition holds with equality. Thus, there exists a tipping point degree of wage theft which can induce the worker to report or not report accordingly. Formally, there exists some t^* such that:

$$p\gamma + (1 - \theta)(1 + \tau)pt^* = \theta[q(1 + \tau)w' + (1 - q)b + (1 + \tau)(w - t^*)] \quad (3)$$

which can be solved for the expression of t^* :

$$t^* = \frac{\theta[q(t + \tau)w' + (1 - q)b + (1 + \tau)w] - p\gamma}{(1 + \tau)[(1 - \theta)p + \theta]} \quad (4)$$

This value is the indifference point for the worker: at a given level of wage theft, t , the decision to report boils down to whether that theft is above or below this indifference value. If $t > t^*$, the worker should report, and if $t < t^*$, they should not.

To understand how changes in the tax structure may affect reporting, consider now a mass of identical workers at identical wages facing a distribution of theft levels. If t^* rises, then some fraction of those workers will be pushed from a position where they prefer reporting into a position where they prefer not reporting, as their respective t falls below the threshold. In other words, a rise in t^* predicts a fall in reporting.

To assess whether taxes, τ , in this model will increase t^* , the derivative of the previous expression can be taken with respect to τ . While the algebra is tedious, the sign of the resulting expression can be condensed to the sign of the following term:

$$p\gamma - \theta(1 - q)b \quad (5)$$

This states that if the expected payout from the report itself (backwages plus possible damages) is greater than the expected welfare of being fired and not finding a new job and receiving unemployment benefits, then the derivative is positive. While it is not immediately clear which direction this inequality runs, it seems that for reasonable values of the involved parameters (particularly in the context of low-income workers and the high success rate of investigations), the effect should be positive.

This is the key prediction of the model which will be tested in this paper: an increase in cash transfers via the EITC should be associated with a decrease in reporting of wage theft.

III Data

This analysis requires three data components: (1) wage theft reports, (2) state EITC rates, and (3) relevant control variables. This section summarizes the sources used for these data, how they were assembled into final sample, and presents summary statistics of that sample in Table 1.

III.A Wage Theft Data

The most-used archive of wage theft reports comes from the US Wage & Hour Division, who maintain thorough documentation of each complaint they receive. Their data extends back to 2005, and includes several variables required for this analysis: address of each complaine, the NAICS code of their employer, the amounts claimed and awarded, and more. Unfortunately, there are also two issues with using this data to measure the desired outcome of report frequency.

Firstly, this set contains two kinds of cases: cases where an employee complained to the government for assistance, and cases where the federal government proactively investigated a firm for potential wage theft. For the purposes of this analysis only the first case type is interesting, the latter unhelpful. The two kinds are not distinguished in the data, creating a nonclassical measurement error issue.

Secondly, workers in the US have two avenues by which to seek redress of wage grievances: the federal DoL and their local state agency. Thus, not only does the federal file measure just a subset of all reports, it is likely a selected sample: individuals in states with effective local complaint management may be more likely to report there than in states with less effective local management, who may prefer the federal option.

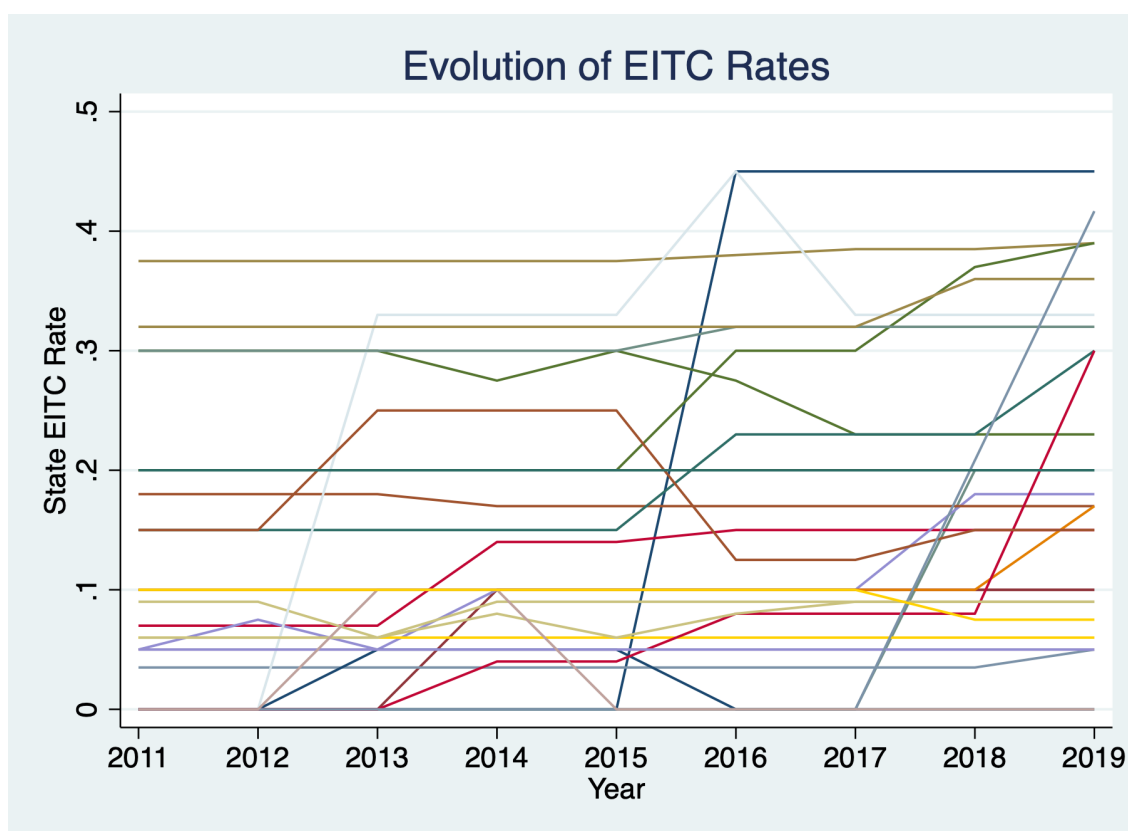
III.B State EITC Data

The explanatory variable of interest, state EITC matching rates, were sourced from the Tax Policy Center at the Urban Institute & Brookings Institution (Tax Policy Center 2023), with gaps filled in using the NBER TAXSIM calculator (Shapiro 2019). In most states, a single

matching rate is provided, meaning that the state matches that percentage of the federal EITC; however, in some states there are different rates depending on factors like number of children. These nuances necessitate some interpretive choices, which I explain in more detail in Appendix A.

Furthermore in Appendix A, I re-run my main analysis using the statutory rates themselves, with no interpretation or adjustment. The resulting estimates are highly similar to my primary analysis, and have similar significance values, giving confidence that my interpretation of those nuanced rates is not skewing the results in a particular direction. The evolution of rates under my interpretation are summarized in Figure 1 (excluding those states with no matching at any point for simplicity).

Figure 1: Evolution of Rates



III.C Control Data

With the outcome variable and explanatory variable in hand, I then incorporated a number of relevant control variables

Tax Return Statistics

Detailed information on the tax returns filed in each ZIP code are available at the IRS Statistics of Income archive (Internal Revenue Service 2022). In particular, I pulled the total number of tax returns and the share of returns which claimed an EITC benefit for inclusion in my analysis.

Business Patterns

The Census County Business Patterns archive includes detailed data at the ZIP code level (U.S. Census 2023b). This includes the number of establishments, the employment share of firms with 50 employees or less, and the share of establishments in several key wage-theft-prone industries: agriculture, construction, and food.

Demographics

Demographic control variables were taken from the American Community Survey Five-Year Data, including variables for race, education, nativity, poverty, and public assistance (U.S. Census 2023a). There are two variables which were not available from the ACS, namely the population density and unemployment rates at the ZIP level. These are available at five-year frequency from the National Neighborhood Data Archive, and I use them as follows: the estimate of 2008-2012 applied over those years, for 2013-2015 applied to those years, and for 2016-2020 applied to those years (Melendez et al. 2023).

State-Level Variables

Two key variables were not available at the ZIP code level, so I used the available state-year level data. State minimum wages in 2020 dollar equivalents were sourced from the Wage & Hour Division (US Department of Labor Wage & Hour Division 2023).³ Unionization rates

³My thanks to Joe Lisle who makes these data more easily available through a compilation available here:

were sourced from Unionstats (Macpherson and Hirsch 2023).

III.D Sample Construction

The various datasets were merged on the ZIP-code year basis, then the sample was trimmed in various ways to avoid outliers driving the results. Firstly, I restricted to those ZIP codes which correspond to the fifty US states, and exclude other ZIP codes for territories, Washington DC, or other administrative units. Second, ZIP codes which at any point reported fewer than 250 tax returns were dropped across all years, as tax information was not reported for any ZIP below this threshold. Third, I drop any ZIP code which at any point reported more than 10 wage theft reports, to avoid my results being driven by these outliers. They account for less than one half percent of the overall sample.

Temporally I restrict my analysis to run from 2011-2019, the former being the first year many of the relevant control variables are available, and the latter cutoff allowing me to avoid the years of the COVID-19 pandemic. These years display radically different wage theft behaviors, and are subject to a host of other concerns that they may otherwise be different in unaccountable ways.

The final sample contains 24,499 ZIP codes, most of which are observed in all nine years 2011-2019 (for a total of 216,652 ZIP-year observations). This sample and the variables used in my analysis are summarized in Table 1.

<https://github.com/Lislejoem/Minimum-Wage-by-State-1968-to-2020/tree/master>.

Table 1: Summary Statistics

	Mean	Std. Dev.	Min	Max
Wage Theft Reports	0.59	1.22	0.00	10.00
State EITC Rate	0.10	0.16	0.00	0.85
State Unionization Rate	0.12	0.05	0.03	0.26
State Minimum Wage	8.37	1.48	2.02	13.66
Tax Returns	5,695	6,917	250	59,000
Share Claiming EITC	0.33	0.12	0.00	0.83
Share Below Poverty	0.10	0.08	0.00	1.00
Share Black	0.09	0.16	0.00	1.00
Share <HS Education	0.13	0.09	0.00	0.92
Share Noncitizen	0.04	0.06	0.00	0.61
Share Receiving Public Assistance	0.25	0.17	0.00	1.00
Share Establishments in Construction	0.13	0.10	0.00	1.00
Share Establishments in Agriculture	0.01	0.03	0.00	1.00
Share Establishments in Food Services	0.08	0.06	0.00	1.00
Share Employment in <50 Worker Establishments	0.97	0.03	0.00	1.00
Population Density	1,515	5,481	0.05	159,897
Unemployment Rate	0.05	0.03	0.00	0.74

Note: Variables are at the level of the ZIP-code-year unless specified as the state level.

IV Empirical Approach

IV.A Primary Outcome - Reporting

Given the nature of the outcome of interest (the number of wage theft reports in a given ZIP code-year) it is appropriate to employ a count regression analysis. The distribution of report counts across ZIP-years (seen below in Figure 2) is positively skewed with a large proportion of zeroes - I will accordingly use a Poisson regression as my preferred specification. Note that while just over 71% of observations have no reports, less than a quarter of ZIP codes never experience a report in any year of the observation period.

Figure 2: Histogram of Reports



Even with the correct regression specification, the relationship between state EITC rates and report frequencies may be corrupted by endogeneity, by primarily omitted variables and reverse causality. Addressing these in turn, I believe that my set of covariates adequately control for the relevant variables. The relative sizes of ZIP codes are accounted for through

the number of establishments and the number of tax returns filed. The labor market conditions are measured through the ZIP unemployment rate, poverty level, public assistance engagement, and the state unionization rate. Furthermore, the industrial makeup of the ZIP is controlled for through the establishment shares of key wage theft industries (agriculture, construction, food service) and the share of employment at smaller (<50 worker) firms. On top of these detailed controls, I account for unobserved differences in state environments through state fixed effects, as well as through national shocks through year effects. I believe this comprehensive set of controls is sufficient to ensure that the estimated effect of state EITC rates is free of contamination from omitted variables.

With regards to reverse causality, it seems highly unlikely that the setting of state EITC generosity is done in consideration of the frequency of wage theft in that state's ZIP codes. These rates are not only set at the state level, far removed from any given ZIP code, but are almost always considered in the context of broader tax reforms, which are subject to numerous higher-order concerns such as fairness, budgetary needs, and economic conditions. That the occurrence of wage theft at a particular level would be a contributing factor to these decisions seems implausible.

To estimate the effect of state EITC rates via maximum likelihood, I assume the following conditional fixed effects Poisson model:

$$E[R_{zy} | \delta_s, \gamma_y, SEITC_{s(z)y}, X_{zy}] = \exp(\delta_s + \gamma_y + \beta \cdot SEITC_{s(z)y} + \Theta X_{zy}) \quad (6)$$

where R_{zy} is the number of wage theft reports in ZIP code z in year y , δ_s are state fixed effects, γ_y are year effects, X_{yz} are the various control variables mentioned, and β is the coefficient of interest on the variable $SEITC_{s(z)y}$, the state EITC rate for the state s of ZIP code z in year y . Estimation of Equation 6 by quasi-maximum likelihood with fixed effects should lead to consistent estimates (Wooldridge 1999).

IV.B Secondary Outcomes - Intensive Measures

The results of the above estimation would not alone be sufficient to claim that there is a negative effect of transfers and reporting - it could be the case that transfers reduce theft frequency itself, mechanically generating a decline in reporting. To argue against this alternative mechanism, I analyze several outcomes which characterize employer behavior in wage theft. If the alternative mechanism is truly driving the observed change in reporting, one would expect to also see changes in these measures. I present each in turn for completeness.

Empty Reports

When a wage theft report is documented with no findings - that is, no backwages owed, no legal violations found, and no civil monetary penalties assessed - I refer to that as an “empty” report. While the exact nature of these is uncertain in the Wage & Hour Division data, I believe it not unreasonable to treat them as reports where no crime was identified. If employers were responding to transfers via an alternative mechanism and engaging in less wage theft, I would predict a positive effect of transfers on the count of empty reports, as there would be less wage theft to find. In contrast, if the effect of transfers on reporting is driven by the worker-side mechanism, then we would expect no change. I estimate the effect via similar model to Equation 6.

As an additional robustness, I estimate this effect via a probit using the share of reports which are empty as the outcome.

Repeat Offenders

Many firms accused of wage theft have been previously investigated for earlier reports. If employers were changing their behavior in response to changes in transfers, it seems logical that there would be a decrease in the number of repeat offenders who are reported, as fewer firms would see the crime as worth the risk. I investigate this dimension of employer behavior via a similar count model. I define a repeat offender as a firm within a given ZIP code which had a report of wage crime in the previous year.⁴

⁴I restrict the definition to include reports which were not within 90 days of the previous report, as this is

Total Violations

In a wage theft investigation, the authorities record each violation of the laws the Wage & Hour Division is responsible for enforcing. I offer this as a rough proxy for the severity / intensity of the wage theft taking place. In particular, if employers respond to transfer changes by reducing their wage theft, out of greater fear of being reported, it seems highly likely they would seek to reduce the number of laws they are breaking in the process. I use another Poisson model for estimating the effect of transfers on violations.

Civil Monetary Penalties

In addition to ordering repayment of backwages, wage theft investigations can lead to civil penalties for the employer in accordance with the laws they have violated. By similar thinking to penalties, this proxies for the intensive margin of wage theft, and should be decreasing if employers change their behavior. Unfortunately, this variable is not a count, but a continuous variable, and so I cannot use a Poisson model.

Instead, I use a Heckman selection model to estimate the effect of transfers on civil penalties. The sample of reports wherein penalties are found are certainly a selected sample among wage theft reports, thus estimating a simple linear model would produce a biased estimate. For identification, I omit the following variables from the penalties estimation but include them in the binary step: State Minimum Wage, Share of ZIP Code Claiming EITC, and share of the population earning over 400% of the poverty line. I believe each of these variables likely has an effect on whether or not penalties are assessed on a report, but do not influence in turn the amount of those penalties.

likely not a true "repeat" but rather a secondary report filed as a follow-up to the original, and not reflective of truly repeating the same crime twice.

V Results

V.A Primary Outcome - Reporting

Poisson modelling of wage theft reporting is summarized in Table 2. Beginning with a naive regression using the single explanatory variable of state EITC rate, Column 1 estimates a statistically significant negative effect, with a coefficient of -1.153. The sign and significance of this estimate is robust to the inclusion of two-way fixed effects for state and year in Column 2 as well as a battery of controls in Column 3 (though the magnitude falls to -0.440) . To further assess the strength of this result, Column 4 uses ZIP code fixed effects in place of state fixed effects, controlling for greater unobserved heterogeneity, producing highly similar results (coefficient -0.372). Finally, Column 5 estimates the effect in a binary probit model, where the outcome is whether any wage theft reports occurred in a given ZIP-year. Here as well I find a significant negative effect.⁵

While not the focus of this analysis, it is worth noting the coefficient on the ZIP code unemployment rate, for comparison with previous work like Dahl and Knepper (2021) and Boone et al. (2011). They use unemployment conditions as the main variation of interest in explaining changes in reporting behavior, finding that worsening unemployment conditions have a suppressive effect on reporting of workplace crime. In line with those papers, I find that an increase in the unemployment rate has a negative correlation with reporting, with a sizeable effect estimate in Columns 3 and 5.⁶

⁵While I have clustered my standard errors at the level of the state-year to allow for correlation of the errors within those units, this in turn creates an issue of heterogeneously sized clusters. I address concerns related to this issue in Appendix Section B. In brief, the results presented here are robust to bootstrapping inference through the wild score cluster bootstrap, as recommended by MacKinnon et al. (2023).

⁶While the effect becomes insignificant in Column 4 with ZIP code fixed effects, it seems likely that here the ZIP effects are absorbing significant unobserved heterogeneity related to labor market conditions with a similar effect.

Table 2: Wage Theft Report Results

	(1)	(2)	(3)	(4)	(5)
	Naive Poisson	TWFE Poisson	Complete Poisson	ZIP FE Poisson	Probit
State EITC Rate	-1.153*** (0.260)	-0.373*** (0.135)	-0.440*** (0.154)	-0.372** (0.150)	-0.204** (0.102)
State Unionization Rate			-0.857 (0.905)	-0.517 (0.894)	-0.728 (0.665)
State Minimum Wage			0.006 (0.015)	0.001 (0.015)	-0.010 (0.010)
Log Tax Returns			0.872*** (0.009)	0.207** (0.094)	0.631*** (0.006)
Share Claiming EITC			-0.822*** (0.154)	0.092 (0.240)	-0.294*** (0.094)
Share Below Poverty			1.543*** (0.112)	0.365** (0.147)	0.913*** (0.090)
Share Black			-0.078 (0.058)	0.480** (0.201)	0.127*** (0.047)
Share <HS Education			-0.791*** (0.114)	-0.227 (0.204)	-0.393*** (0.091)
Share Noncitizen			0.991*** (0.137)	0.106 (0.280)	1.532*** (0.130)
Share on Pub. Assist.			0.547*** (0.063)	0.047 (0.073)	0.122*** (0.044)
Share Est in Con			-2.094*** (0.106)	0.443 (0.294)	-0.896*** (0.064)
Share Est in Ag			-1.160*** (0.406)	0.200 (0.763)	-0.063 (0.170)
Share Est in Food Svc.			0.786*** (0.171)	0.556* (0.304)	0.912*** (0.097)
Share Emp <50			-7.199*** (0.477)	-1.178*** (0.359)	-5.289*** (0.213)
Log Population Density			-0.013* (0.006)	0.038 (0.037)	-0.003 (0.005)
Unemployment Rate			-1.587*** (0.333)	-0.098 (0.305)	-0.822*** (0.213)
State & Year FE		✓	✓	ZIP & Year	✓
R ²	0.00	0.06	0.32	0.31	0.31
N	216,652	216,652	216,652	143,830	216,652

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

V.B Secondary Outcomes - Intensive Measures

The results above cannot alone distinguish the two possible mechanisms of worker or employer behavioral change. To address this concern, I offer results of several analyses of the intensive margin of employer wage theft behavior. While none of them on their own is definitive in arguing for the worker-side mechanism, I believe their totality makes a strong argument that employers are not reducing their wage theft behavior in response to work contingent transfer changes.

Beginning in Column 1, the effect of EITC rates on the number of "empty" reports, where no wage theft or other workplace crime is found, is null. A similar null result is estimated for the effect of EITC rates on the share of all reports which are empty in Column 2. These empty reports are difficult to characterize - some may be agency-initiated investigations which fail to find evidence, or they could be "frivolous" reports filed by disgruntled employees. This difficulty notwithstanding, if it was true that employers were decreasing their wage theft behavior in response to rising EITCs, this would suggest a positive coefficient estimate, rather than the observed negative (though insignificant) estimate in Column 1.

Another null effect is found for the effect of EITC rates on repeat offender counts in Column 3. Again, it seems likely that if employers were reducing wage theft behavior in response to a shift in transfer policy, there would be fewer repeat offenders counted among those committing wage theft. The lack of effect further disputes this interpretation.

The effect on total labor violations in Column 4 is less straightforward to interpret. The negative coefficient (statistically significant at the 10% level) is indeed in line with the prediction of a reduced wage theft mechanism. If employers are reducing the intensity of wage theft behavior, one would expect fewer violations being found in wage theft investigations, and this estimate agrees; however, the significance of this estimate is not robust to the wild score cluster bootstrap, as shown in Appendix B. This method is important for inference with my data, due to the high heterogeneity of size among state-year clusters. While my other results remain significant under this increased scrutiny, this one falls away, making it less likely that the total number of violations is truly changing in response to state EITC

rates.

The effect for civil monetary penalties in Column 5 is also less straightforward to interpret; however, it is certainly not the case that this result supports an employer behavior adjustment mechanism. The estimated effect is positive and highly statistically significant (even when bootstrapping in Appendix B. This suggests that more penalties are assessed on wage thefts reported at higher levels of state EITC. This could mean in turn that the kind of wage theft which is reported at higher EITC levels is more severe / intense (more worthy of being penalized) than at lower EITC rates.

Returning to the model in Section II, this result agrees with the intuition presented. At higher EITC levels, workers are willing to "put up" with greater / more intense wage theft, as the value of work rises such that the risk of retaliatory firing is not worth running. Accordingly, those workers who do still choose to report are likely experiencing more severe wage theft, as for them the benefits of reporting still outweigh the potential costs. While the identification assumptions used in this Heckman selection model are imperfect, the result of a positive estimate is certainly not evidence for reduced employer wage theft intensity. This result is another mark against the idea that the decreased reporting found above is being driven by changes in employer behavior, but rather that it reflects a decrease in reporting propensities among workers. In particular, this result highlights a potential shift in the nature of the reports being filed, towards greater intensity, which is in line with model predictions. A similar finding is reported in Dahl and Knepper (2021), that the kinds of reports filed become more severe as outside options become less attractive relative to remaining employed.

Table 3: Intensive Margin Results

	(1)	(2)	(3)	(4)	(5)
	Empty Reports	Empty Share	Repeat Offenders	Total Vltns.	Civil Penalties
	Poisson	Probit	Poisson	Poisson	Heckman
State EITC Rate	-0.075 (0.189)	0.034 (0.137)	-0.659 (0.445)	-0.186* (0.108)	1.050*** (0.190)
State Unionization Rate	-1.503 (1.441)	-0.889 (1.005)	2.640 (4.473)	0.128 (0.919)	0.258 (2.226)
State Minimum Wage	-0.000 (0.020)	-0.005 (0.014)	-0.023 (0.060)	0.012 (0.014)	
Log Tax Returns	0.217*** (0.013)	0.111*** (0.009)	0.800*** (0.048)	0.257*** (0.010)	-0.216*** (0.025)
Share Claiming EITC	-0.179 (0.159)	-0.031 (0.121)	-2.007*** (0.711)	0.023 (0.127)	
Share Below Poverty	1.015*** (0.198)	0.596*** (0.149)	2.241** (0.910)	0.323* (0.168)	-0.585 (0.412)
Share Black	-0.198*** (0.068)	-0.124** (0.053)	0.477** (0.238)	-0.079 (0.056)	0.269* (0.141)
Share <HS Education	-0.891*** (0.170)	-0.591*** (0.129)	-0.008 (0.628)	-0.488*** (0.133)	0.362 (0.329)
Share Noncitizen	0.778*** (0.202)	0.607*** (0.158)	1.455** (0.702)	1.250*** (0.145)	0.266 (0.368)
Share on Pub Assist.	0.124 (0.109)	0.039 (0.078)	0.832* (0.426)	0.089 (0.086)	-0.422** (0.172)
Share Est in Con	0.014 (0.168)	0.210* (0.116)	-2.525*** (0.615)	-1.042*** (0.141)	0.401 (0.278)
Share Est in Ag	0.002 (0.804)	-0.392 (0.524)	1.098 (1.424)	0.071 (0.567)	0.403 (0.940)
Share Est in Food Svc.	-0.631** (0.253)	-0.328* (0.179)	0.564 (0.763)	-0.019 (0.201)	0.041 (0.344)
Share Emp <50	-4.887*** (0.266)	-3.032*** (0.208)	-9.278*** (0.827)	-4.725*** (0.241)	2.772*** (0.614)
Log Population Density	-0.012 (0.009)	-0.002 (0.006)	-0.033 (0.030)	-0.004 (0.006)	0.009 (0.019)
Unemployment Rate	-0.362 (0.476)	-0.208 (0.325)	-1.945 (1.945)	-0.592* (0.343)	0.347 (0.647)
State & Year FE	✓	✓	✓	✓	✓
N	63,749	63,749	216,652	63,749	63,749

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

VI Discussion

Having established the statistical significance of the relationship between transfers and wage theft, it is useful to translate the estimated effect into economic significance. In my preferred specification (Table 2, Column 3), the estimated coefficient of -0.44 implies the following: a state which went from zero EITC match to matching 100% of the federal credit would see a reduction in wage theft reporting of 44%. Considering this effect at the level of the ZIP code is not so interesting, given the average ZIP code in my sample experiences only 0.5 reports in a year; however, scaling this up to a state-year average of 600 reports, this would imply a fall to 324 reports per year. Factoring in the average amount of theft per report, and this would imply an additional \$2.9 million dollars of unreported wage theft annually.

On its own, this effect may still not seem terribly significant - most states are unlikely to make such dramatic changes in their EITC match - however, this effect adds to the compelling set of results in the literature that identify the ways worker power affects workplace crime and safety. In this broader context, cash transfers act as something which increases the value of work - through effectively increased income. As shown in work like Boone et al. (2011) and Dahl and Knepper (2021), when the value of work relative to unemployment increases due to worsening unemployment prospects, reporting of workplace crimes goes less reported (safety violations and sexual assault, respectively). Here, that same ratio of job value to unemployment increases, but due to the denominator rather than denominator. When one considers the numerous levers at the government's disposal to affect this tradeoff, and the many crimes and costs associated with it, it creates a strong lens for additional policy benefit analysis.⁷

Furthermore, this finding emphasizes the importance of retaliatory firing as a key mechanism for underreporting, as shown in the context of safety by Johnson et al. (2023). Greater protection from retaliatory firing would likely also increase the rates of wage theft reporting, and potentially decrease its prevalence as well.

⁷Additionally for this mechanism, consider that my estimates for the effect of the unemployment rate are also consistently negative for reporting, there driving down the value of unemployment prospects and along with it, reporting.

VII Conclusion

Wage theft is an understudied workplace crime, largely due to underreporting. To understand why workers decide to report wage theft so infrequently, this paper investigates one possible factor in that decision: work-contingent government transfers. These transfers have a direct effect on the relative value of remaining employed relative to becoming unemployed, which is the core of a key mechanism in the workplace crime reporting literature. When this relative value increases, workers are less likely to report, as the inherent risk of retaliatory firing becomes more threatening. In line with this mechanism and the literature, I find that increases in state EITC matching are associated with decreased counts of wage theft reports. This finding is robust to a variety of empirical specifications.

To further argue that this decrease is driven by changes in worker behavior (rather than changes in employer wage theft behavior itself), I present several secondary outcomes on employer behavior which do not support the idea of decreased wage theft intensity. These results suggest a focus on firm-side intensive-margin responses in future research. This can help focus additional studies in trying to better identify how firms learn about welfare and perceive employee responses to wage theft changes.

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Appendix

A EITC Rate Definitions

The values used for state EITC rates are discussed in Section III but bear further explanation. While many states use a simple matching rate, several use alternative structures. Here I review those alternatives and how I encoded values from the description:

California

California's credit has a smaller eligible income range than the federal credit, with a maximum which rose over the active years in my sample of analysis. The Tax Policy Center has summarized this divergence in recent years by listing the rate as a 45% match (rather than the statutory 85% figure). I follow this example and use the 45% rate in my set.

Maryland

Maryland employs two separate matching rates, one for refundable credits (used in the baseline) and one for non-refundable at 50%. To approximate the effective rate of this combination, I use the average of the varying statutory rate and 50%.

Minnesota

Minnesota's credit is different from other states in that it is not structured as a percentage of the federal credit. The Tax Policy Center reports an average rate of 33%, which I employ in the baseline. I do not adjust this rate for my preferred set, but simply report this usage for completeness.

New Jersey

New Jersey limits their match to a maximum income of \$20,000. This is roughly half the total federal EITC range, though it is in the "plateau" region of the maximum credit. Accordingly, as an alternative to the statutory rate, I use three-quarters of that rate to reflect the diminished significance.

New York

New York City has its own supplemental 5% EITC. Given that roughly 40% of the state lives in the city, I adjust the statutory rate of 30% up to 32% in my preferred set.

Ohio

Ohio's credit is limited to 50% of liability for state taxable income above \$20,000. I halve Ohio's rates to attempt and reflect this diminished significance.

Oregon

Oregon switched from a single rate for all filers to having a separate rate for filers with children under the age of three in 2017. Accordingly, I adjust the statutory rate of 8% up to 9% as my preferred rate, to reflect that a share of filers are receiving the new separate higher rate of 11%.

Wisconsin

Wisconsin's credit is only available to workers with qualified children, and uses separate rates by number of children, as one (4%), two (11%), and three (34%). To try and match this in an aggregate, I use the 4% figure in the statutory and adjust to 10% in the preferred rate.

To avoid the potential that my results are driven by my interpretation of these rates, Table 4 presents the results of my main estimation using the original statutory rates, free of adjustment or interpretation. The results are highly similar to those found in Table 2, suggesting that these irregularly defined rates are not somehow driving a spurious result.

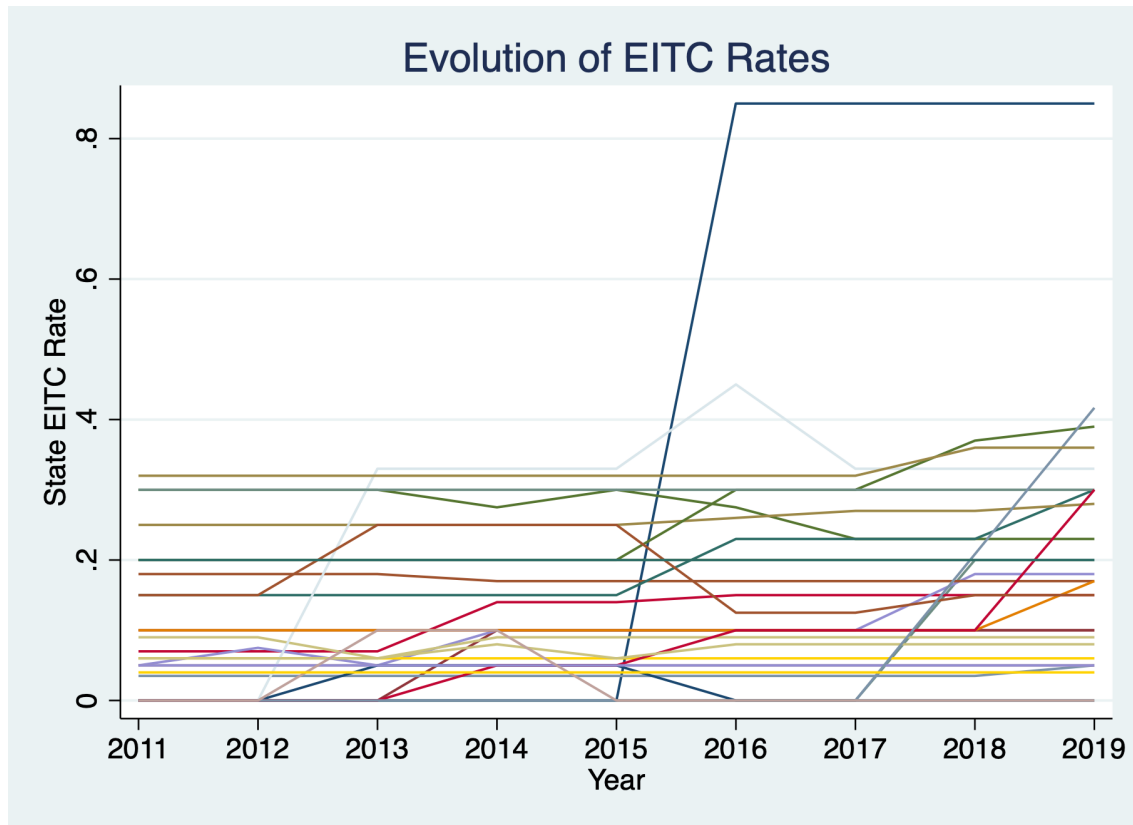
Table 4: Count Results - Statutory Rates

	(1)	(2)	(3)	(4)	(5)
	Naive Poisson	TWFE Poisson	Complete Poisson	ZIP FE Poisson	Probit
State EITC Rate	-0.673*** (0.261)	-0.271*** (0.061)	-0.319*** (0.072)	-0.281*** (0.069)	-0.175*** (0.054)
State Unionization Rate			-0.960 (0.889)	-0.620 (0.880)	-0.799 (0.653)
State Minimum Wage			0.010 (0.014)	0.005 (0.014)	-0.007 (0.010)
Log Tax Returns			0.872*** (0.009)	0.218** (0.094)	0.631*** (0.006)
Share Claiming EITC			-0.824*** (0.154)	0.075 (0.238)	-0.296*** (0.094)
Share Below Poverty			1.539*** (0.112)	0.359** (0.148)	0.911*** (0.090)
Share Black			-0.078 (0.058)	0.463** (0.201)	0.127*** (0.047)
Share <HS Education			-0.791*** (0.114)	-0.236 (0.204)	-0.392*** (0.090)
Share Noncitizen			0.989*** (0.137)	0.045 (0.277)	1.529*** (0.130)
Share on Pub. Assist.			0.551*** (0.063)	0.056 (0.072)	0.124*** (0.044)
Share Est in Con			-2.092*** (0.106)	0.453 (0.294)	-0.895*** (0.064)
Share Est in Ag			-1.158*** (0.406)	0.189 (0.761)	-0.061 (0.170)
Share Est Food Svcs.			0.788*** (0.171)	0.584* (0.302)	0.914*** (0.097)
Share Emp <50			-7.202*** (0.477)	-1.171*** (0.358)	-5.291*** (0.213)
Log Population Density			-0.013* (0.006)	0.037 (0.037)	-0.003 (0.005)
Unemployment Rate			-1.593*** (0.332)	-0.092 (0.303)	-0.821*** (0.213)
State & Year FE		✓	✓	ZIP & Year	✓
R ²	0.00	0.06	0.32	0.31	0.31
N	216,652	216,652	216,652	143,830	216,652

Note: Standard errors are clustered at the state-year level. * = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

Finally, the evolution of the statutory rates can be visualized below in Figure 3. While labelling every state is impractical, in anticipation of interest I note that the state which transitioned from no rate to 85% match in 2016 is California.

Figure 3: Evolution of Rates



B Cluster Size Heterogeneity

While clustering at the state-year level ensures that I have a relatively large number of clusters and allows for what I believe is the appropriate level of error correlation, there is significant size heterogeneity among those clusters. This is due to the variation in number of underlying ZIP codes per state, which vary by an order of magnitude at the extremes (despite most being fairly similar in the several hundreds count). This kind of heterogeneity can lead to over-rejection in hypothesis testing, as the larger clusters will tend to have much higher leverage, meaning that asymptotic theory may not carry through reliably. To overcome this issue, I employ the wild score cluster bootstrap approach, as developed by Kline and Santos (2012) and recommended in this setting by MacKinnon et al. (2023).

Implementing this via the Stata command `boottest` (Roodman et al. 2019), I recreate the key portions of Tables 2 and 3 below with the significance levels calculated accordingly. The results are largely unchanged, giving confidence that the cluster size heterogeneity is not driving inference misleadingly. Note that the ZIP-code level analysis could not be duplicated in bootstrap, as the `boottest` command is incompatible with the relevant high-dimensional fixed effects Poisson command.

Table 5: Count Results - Bootstrapped Significance

	(1)	(2)	(3)	(4)
	Naive Poisson	TWFE Poisson	Complete Poisson	Probit
State EITC Rate	-1.153***	-0.373*	-0.440**	-0.204*
P-value	(0.000)	(0.051)	(0.0220)	(0.081)
N	216,652	216,652	216,652	216,652

Note: P-values are presented in parentheses, and were calculated using the wild score cluster bootstrap.

* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.

Table 6: Intensive Margin Results - Bootstrapped Significance

	(1)	(2)	(3)	(4)	(5)
	Empty Reports	Empty Share	Repeat Offenders	Total Vltns.	Civil Penalties
	Poisson	Probit	Poisson	Poisson	Heckman
State EITC Rate	-0.075	0.034	-0.659	-0.186	1.050**
P-value	(0.691)	(0.818)	(0.204)	(0.116)	(0.011)
N	63,749	63,749	216,652	63,749	63,749

Note: P-values are presented in parentheses, and were calculated using the wild score cluster bootstrap.

* = $p < 0.1$, ** = $p < 0.05$, *** = $p < 0.01$.